

14 (a) Expand and simplify $7x(3x+2)(2x-5)$

~~340~~

$$7x[3x(2x-5) + 2(2x-5)]$$

$$7x[6x^2 - 15x + 4x - 10]$$

$$7x[6x^2 - 11x - 10]$$

$$= \underline{42x^3 - 77x^2 - 70x}$$

$$42x^3 - 77x^2 - 70x$$

(3)

(b) Solve $\frac{7x-9}{7x-2y} + \frac{5-2y}{7-2y} = 5$

Show clear algebraic working.

$$1 \times \frac{63 + 10y}{14y} = 5 \times 14y$$

$$63 + 10y = 70y$$

$$63 = 70y - 10y$$

$$\frac{63}{60} = \frac{60y}{60}$$

$$\underline{y = 1.05}$$

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(3)

(Total for Question 14 is 6 marks)

